

(a) We can start by algebraically eliminating the λ term. Write expressions for u and v in terms of $h_1, \dots, h_8$, and x, y . (hint: division)

$$\begin{aligned}\lambda u &= h_1 x + h_2 y + h_3 \\ \lambda v &= h_4 x + h_5 y + h_6 \\ \lambda &= h_7 x + h_8 y + 1\end{aligned}$$

$$\begin{aligned}u &= \frac{h_1 x + h_2 y + h_3}{h_7 x + h_8 y + 1} \\ v &= \frac{h_4 x + h_5 y + h_6}{h_7 x + h_8 y + 1}\end{aligned}$$

(b) Convert your two equations from Part (a) into linear equations in the form: $(\dots)h_1 + (\dots)h_2 + \dots + (\dots)h_8 = (\dots)$. Why is this useful?

$$h_1 x + h_2 y + h_3 - h_7 x u - h_8 y u = u$$

$$h_4 x + h_5 y + h_6 - h_7 x v - h_8 y v = v$$

(c) What a_1^T , a_2^T , b_1 , and b_2 values does this correspondence pair produce?

$$a_1^T = [x \quad y \quad 1 \quad 0 \quad 0 \quad 0 \quad -xu \quad -yu]$$

$$a_2^T = [0 \quad 0 \quad 0 \quad x \quad y \quad 1 \quad -xv \quad -yv]$$

$$b_1 = u \quad b_2 = v$$

Problem 2.2: Multiple Correspondences

(a) Suppose we have multiple correspondence pairs: $(x_1, y_1) \rightarrow (u_1, v_1)$, $(x_2, y_2) \rightarrow (u_2, v_2)$, $(x_3, y_3) \rightarrow (u_3, v_3)$, $(x_4, y_4) \rightarrow (u_4, v_4)$. Fill in the blanks in this matrix system for 4 correspondence pairs:

$$\begin{bmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 & -xu & -yu \\ 0 & 0 & 0 & x_1 & y_1 & 1 & -x_1 v & -y_1 v \\ x_2 & y_2 & 1 & 0 & 0 & 0 & -x_2 u & -y_2 u \\ 0 & 0 & 0 & x_2 & y_2 & 1 & -x_2 v & -y_2 v \\ x_3 & y_3 & 1 & 0 & 0 & 0 & -x_3 u & -y_3 u \\ 0 & 0 & 0 & x_3 & y_3 & 1 & -x_3 v & -y_3 v \\ x_4 & y_4 & 1 & 0 & 0 & 0 & -x_4 u & -y_4 u \\ 0 & 0 & 0 & x_4 & y_4 & 1 & -x_4 v & -y_4 v \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \\ h_5 \\ h_6 \\ h_7 \\ h_8 \end{bmatrix} = \begin{bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \end{bmatrix}$$